Deriving $c(x)$: the control-authority ceiling of an SSA safety constraint

SPARK goal-insertion notes

Setup and notation

symbol	meaning
$x \in \mathbb{R}^n$	robot configuration (joint positions; the safety-index state)
$u \in \mathbb{R}^m$	control input
$\mathcal{U} = \{u : \lvert u_k \rvert \le u_{lim,k}\}$	actuator box ($= \prod_k [-u_{lim,k},\, u_{lim,k}]$)
$\dot{x} = f(x) + G(x)\, u$	control-affine dynamics; $G(x)$ = actuation map
$\phi(x)$	safety index (one constraint); safe set $\{\phi \le 0\}$, danger $\{\phi > 0\}$
$\eta > 0$	SSA margin rate: enforce $\dot{\phi} \le -\eta$ when $\phi \ge 0$
n	unit contact normal (the distance-decreasing / escape direction)
$J(x)$	Jacobian of the contacting body point, $\mathbb{R}^{3\times n}$

We derive $c(x)$, the maximum rate at which the actuators can drive ϕ down, and show the single-active-constraint QP is feasible iff $\eta \le c(x)$ (first order).

Step 1 — the time derivative of the safety index

Differentiating ϕ along the dynamics and using control-affineness,

$$\dot{\phi} = \nabla\phi(x)^\top \dot{x} = \nabla\phi^\top\big(f(x) + G(x)u\big) = \underbrace{\nabla\phi^\top f}_{=:\,L_f\phi} + \underbrace{\nabla\phi^\top G}_{=:\,L_g\phi}\; u.$$

So $\dot{\phi} = L_f\phi + L_g\phi\, u$, where $L_f\phi$ is the drift (uncontrolled rate) and $L_g\phi \in \mathbb{R}^{1\times m}$ is the control gain of the index.

For a collision index, ϕ depends on x only through the contact-point position $p(x)$, with Cartesian gradient equal to the normal n. The chain rule $\nabla_x\phi = J(x)^\top n$ then gives

$$\boxed{L_g\phi(x) = n^\top J(x)\, G(x)} \qquad (\in \mathbb{R}^{1\times m}).$$

Step 2 — feasibility of the safety constraint

The SSA QP requires a control in the box that satisfies the safety inequality:

$$\exists\, u \in \mathcal{U}: \quad L_f\phi + L_g\phi\, u \le -\eta.$$

Such a u exists iff the most negative achievable $\dot\phi$ already meets the bound:

$$\min_{u\in\mathcal{U}} \big(L_f\phi + L_g\phi\, u\big) \;\le\; -\eta \quad \Longleftrightarrow \quad L_f\phi + \min_{u\in\mathcal{U}} L_g\phi\, u \;\le\; -\eta.$$

Step 3 — minimizing a linear form over the box

$L_g\phi\, u \;=\; \sum_k (L_g\phi)_k\, u_k$ is separable across coordinates, so it is minimized coordinate-wise. Pushing each u_k to the box corner that makes its term most negative,

$$u_k^\star = -\,\mathrm{sign}\big((L_g\phi)_k\big)\, u_{lim,k} \quad \Longrightarrow \quad (L_g\phi)_k\, u_k^\star = -\,|(L_g\phi)_k|\, u_{lim,k},$$

hence

$$\min_{u\in\mathcal{U}} L_g\phi\, u \;=\; -\sum_{k=1}^{m} u_{lim,k}\, \big|(L_g\phi)_k\big|.$$

Step 4 — define $c(x)$

Define the control-authority ceiling as the maximum achievable decrease rate:

$$c(x) \;:=\; \max_{u\in\mathcal{U}} \big(-L_g\phi\, u\big) \;=\; -\min_{u\in\mathcal{U}} L_g\phi\, u \;=\; \sum_{k=1}^{m} u_{lim,k}\, \big|(L_g\phi)_k\big|.$$

Substituting into Step 2, the feasibility condition becomes

$$L_f\phi - c(x) \le -\eta \quad \Longleftrightarrow \quad \boxed{c(x) \;\ge\; \eta + L_f\phi}.$$

First-order / velocity control ($f \equiv 0 \Rightarrow L_f\phi = 0$): this collapses to

$$\boxed{\text{feasible} \;\Longleftrightarrow\; c(x) \ge \eta\,, \qquad c(x) = \sum_k u_{lim,k}\, \big|(L_g\phi)_k\big| = \big\|\mathrm{diag}(u_{lim})\, L_g\phi^\top\big\|_1.}$$

Step 5 — closed form in kinematic terms

Inserting $L_g\phi = n^\top J(x)\, G(x)$ from Step 1:

$$c(x) \;=\; \sum_{k=1}^{m} u_{lim,k} \left| \big[\, n^\top J(x)\, G(x)\,\big]_k \right| \;=\; \left\| \mathrm{diag}(u_{lim})\, \big(n^\top J(x) G(x)\big)^\top \right\|_1 .$$

It is the actuator-limit-weighted ℓ_1 norm of the safety gradient in control space.

Step 6 — geometric reading (support function / velocity zonotope)

$c(x)$ is the support function of the box evaluated in the direction $-L_g\phi$: $c(x) = h_{\mathcal{U}}(-L_g\phi)$. Equivalently, as u sweeps $\mathcal{U}$ the contact point's velocity $v = J(x)G(x)\,u$ traces a zonotope $\mathcal{Z} = JG\,\mathcal{U}$, and since $\dot\phi = n^\top v$,

$$c(x) \;=\; \max_{v\in\mathcal{Z}} \big(-n^\top v \big) \;=\; h_{\mathcal{Z}}(-n)$$

— how far the reachable-velocity set extends in the escape direction $-n$. The reachable $\dot\phi$ is exactly the interval $[-c(x),\, +c(x)]$ (symmetric box), so $c(x)$ is its half-width.

Step 7 — special cases

- Ball-limited actuation $\mathcal{U} = \{\|u\|_2 \le r\}$: $h_{\mathcal{U}}(y) = r\|y\|_2$, so $c(x) = r\,\|L_g\phi\|_2$ (same gradient, ℓ_2 instead of weighted-ℓ_1 — the norm comes from the limit-set geometry).
- Per-contact reduction: $J(x)$ has zero columns for joints distal to the contacting body, so $(L_g\phi)_k = 0$ there. Only joints proximal to the contact contribute to $c(x)$ — the effective dimension is the proximal sub-chain.

Step 8 — the degenerate case $c(x) = 0$

Since $c(x)$ is a sum of non-negative terms,

$$c(x) = 0 \iff L_g\phi = n^\top J(x)G(x) = \mathbf{0} \iff \big(J(x)G(x)\big)^\top n = \mathbf{0} \iff n \perp \mathrm{Im}\big(J(x)G(x)\big).$$

The escape direction is orthogonal to the entire reachable-velocity subspace: the body can move only tangent to the obstacle, never separate from it. This is a controllability singularity of the constraint ($\partial\dot\phi/\partial u = 0$) — not actuator saturation, not a closed-loop equilibrium. At such x no $\eta > 0$ is feasible.

Step 9 — extension to multiple active constraints

The scalar test holds for one active constraint. With several active $i \in \mathcal{A}$, write $a_i := L_g\phi_i^\top \in \mathbb{R}^m$ and $\beta_i := \eta_i + L_f\phi_i$. The QP is feasible iff

$$\exists\, u \in \mathcal{U} :\ a_i^\top u + \beta_i \le 0\ \ \forall i \quad \Longleftrightarrow \quad \mu(x) := \min_{u\in\mathcal{U}}\ \max_{i\in\mathcal{A}} \big(a_i^\top u + \beta_i\big)\ \le\ 0.$$

$\mu(x)$ is the collective achievable margin — the smallest the most-violated constraint can be driven over the box (a convex piecewise-linear minimization, i.e. an LP).

Dual / aggregated form. Lift $\max_i$ to a weight y on the simplex $\Delta = \{y \ge 0,\ \mathbf{1}^\top y = 1\}$ and swap min / max (Sion's theorem: the payoff $\sum_i y_i(a_i^\top u + \beta_i)$ is bilinear; $\mathcal{U}, \Delta$ compact convex):

$$\mu(x) = \max_{y\in\Delta}\min_{u\in\mathcal{U}} \sum_i y_i(a_i^\top u + \beta_i) = \max_{y\in\Delta}\Big[\, y^\top\beta - \underbrace{\textstyle\sum_k u_{lim,k}\big|(\sum_i y_i a_i)_k\big|}_{=:\, c_y(x)}\Big].$$

The inner min reused Step 3. Crucially, $c_y(x) = \big\|\mathrm{diag}(u_{lim})\,\bar{a}(y)^\top\big\|_1$ is the single-constraint ceiling of Step 4 applied to the aggregated normal $\bar{a}(y) = \sum_i y_i\, L_g\phi_i$. Therefore

$$\boxed{\text{feasible} \iff \forall\, y \in \Delta :\ c_y(x)\ \ge\ \sum_i y_i(\eta_i + L_f\phi_i)\ .}$$

For every nonnegative blend of the active constraints, the aggregated control authority must dominate the aggregated demand. Infeasibility is certified by one witness:

$$\exists\, y^\star \ge 0 : \quad c_{y^\star}(x)\ <\ \sum_i y_i^\star(\eta_i + L_f\phi_i),$$

the pincer: choose $y^\star$ so the normals partially cancel, $\bar{a}(y^\star) = \sum_i y_i^\star L_g\phi_i \approx \mathbf{0}$ (hence $c_{y^\star} \approx 0$), while the demands add, $\sum_i y_i^\star(\eta_i + L_f\phi_i) > 0$. In the limit this is the Farkas certificate $\sum_i y_i L_g\phi_i = \mathbf{0},\ \sum_i y_i(\eta_i + L_f\phi_i) < 0$.

What changed vs. the single-constraint case.

- $|\mathcal{A}| = 1$: $\Delta = \{1\}$, $\bar{a} = a_1$, recovering $c(x) \ge \eta + L_f\phi$ (Step 4).
- Per-constraint test is only necessary: $\eta_i \le c_i(x)\ \forall i$ is the boxed condition at the vertices $y = e_i$; an interior $y^\star$ can still fail it. So no single scalar ceiling suffices — conflict lives in the interior of Δ.
- No closed form: one linear form over a box decouples coordinate-wise (Step 3); a max of several couples the coordinates, so feasibility needs the LP (the maximization over Δ). The structure is still explicit — the support function of the aggregated normal $\bar{a}(y)$.

Summary

$$\boxed{\begin{gathered} c(x) = \max_{u\in\mathcal{U}} \big(-L_g\phi\, u \big) = \sum_k u_{lim,k} \left| [\, n^\top J(x) G(x)\,]_k \right| = h_{\mathcal{Z}}(-n) \\ \text{feasible} \iff c(x) \;\ge\; \eta + L_f\phi \end{gathered}}$$

- It is the best achievable rate of decrease of the safety index under the actuator limits.
- It depends only on the configuration (via J, G) and contact geometry (via n) — not on velocity, for first-order dynamics.
- An attacker steers the closed loop to a boundary contact minimizing $c(x)$; $c(x) \to 0$ at singularities, so lowering η shrinks but never closes the vulnerable set.